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Connection diagnosis device

Abstract

Embodiments of the present invention relates to a connection diagnosis device for diagnosing connection states between devices. The connection diagnosis device may include: an output end and an input end connected to respective ends of an interlock loop line included in one of the devices; a first voltage varying circuit for outputting a first voltage that is variable according to a first current flowing through a first current path from among a current path for transmitting a diagnosis signal to the output end and a current path between the input end and a ground; and a first impedance varying circuit disposed on the first current path and varying impedance of the first current path according to the first voltage. The first voltage varying circuit and the first impedance varying circuit are operated to maintain the first current at a constant value.

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Background/Summary**CROSS-REFERENCE TO RELATED APPLICATIONS**

(1) This application is a U.S. National Phase Patent Application of International Application Number PCT/KR2021/014313, filed on Oct. 15, 2021, which claims priority of Korean Patent Application Number 10-2020-0134473, filed on Oct. 16, 2020, the entire content of each of which is incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to a connection diagnosis device. More particularly, the present invention relates to a connection diagnosis device for diagnosing a connection state of devices in a high voltage system.

BACKGROUND ART

(3) A hazardous voltage interlock loop (HVIL) circuit diagnoses connection states between devices in a system in which connection defects between devices may generate dangerous accidents like a high voltage system. The HVIL circuit conventionally applies specific signals (e.g., DC or AC voltage signals, current signals, and frequency signals) to an interlock loop line, and verifies the signals received through the interlock loop line to thus diagnose the connection state.

(4) When the devices are assembled, the interlock loop line is sometimes short-circuited to a battery or a ground because of misconnection of lines. When an overcurrent generated by a short circuit of the interlock loop line is supplied to the HVIL circuit, overheat may be generated and the circuit may be damaged.

(5) Therefore, it is needed to design the HVIL circuit so as to suppress generation of heat and prevent the circuit from being damaged when the interlock loop line is short-circuited to the ground or the battery.

DISCLOSURE

(6) The present invention has been made in an effort to provide a connection diagnosis device for minimizing generation of heat according to a short circuit and preventing circuit damages when an interlock loop line is short-circuited.

(7) An embodiment of the present invention provides a connection diagnosis device for diagnosing connection states between devices, including: an output end and an input end connected to respective ends of an interlock loop line included in one of the devices; a first voltage varying circuit for outputting a first voltage that is variable according to a first current flowing through a first current path from among a current path for transmitting a diagnosis signal to the output end and a current path between the input end and a ground; and a first impedance varying circuit disposed on the first current path and varying impedance of the first current path according to the first voltage. The first voltage varying circuit and the first impedance varying circuit may be operated to maintain the first current at a constant value.

(8) The first current path may be a current path between the input end and the ground. The connection diagnosis device may further include: a second voltage varying circuit for outputting a second voltage that is variable according to a second current flowing to a second current path for transmitting a diagnosis signal to the output end; and a second impedance varying circuit disposed on the second current path and varying impedance of the second current path according to the second voltage. The second voltage varying circuit and the second impedance varying circuit may be operated so that the second current maintains a constant value.

(9) The first voltage varying circuit and the first impedance varying circuit may be operated so that the first current maintains a first value. The second voltage varying circuit and the second impedance varying circuit may be operated so that the second current maintains a second value. The first value may be different from the second value.

(10) The first voltage varying circuit and the first impedance varying circuit may be operated so that the first current maintains the first value when the first current belongs to a first current section. The second voltage varying circuit and the second impedance varying circuit may be operated so that the second current maintains the second value when the second current belongs to a second current section. The first current section may be different from the second current section.

(11) The second voltage varying circuit and the second impedance varying circuit may be operated so that the second current maintains the second value while the devices are normally connected.

(12) The first voltage varying circuit and the first impedance varying circuit may be operated so that the first current maintains the first value while the interlock loop line is short-circuited to a positive electrode side of a battery. The second voltage varying circuit and the second impedance varying circuit may be operated so that the second current maintains the second value while the interlock loop line is short-circuited to the ground.

(13) The first voltage varying circuit may include a first resistor disposed on the first current path, and a first shunt regulator element including an anode terminal and a reference terminal connected to respective ends of the first resistor, and a cathode terminal for outputting the first voltage that is variable according to the first current flowing to the first resistor. The first impedance varying circuit may include a first transistor disposed on the first current path, and including a first terminal connected to the input end, a second terminal connected to the first resistor, and a control terminal connected to a cathode terminal of the first shunt regulator element, and a second resistor connected between a node to which the diagnosis signal is input and the control terminal of the first transistor. Impedance of the first transistor may be variable according to the first voltage.

(14) The second voltage varying circuit may include a third resistor disposed on the second current path, and a second shunt regulator element including an anode terminal and a reference terminal connected to respective ends of the third resistor, and a cathode terminal for outputting the second voltage that is variable according to the second current flowing to the third resistor. The second impedance varying circuit may include a second transistor disposed on the second current path, and including a first terminal connected to the node, a second terminal connected to the third resistor, and a control terminal connected to a cathode terminal of the second shunt regulator element, and a fourth resistor connected between the node and the control terminal of the second transistor. Impedance of the second transistor may be variable according to the second voltage. Resistance of

the first resistor may be different from resistance of the third resistor.

(15) The first current path may be a current path between the input end and the ground, and the connection diagnosis device may further include a second voltage varying circuit for outputting a second voltage that is variable according to a second current flowing to a second current path for transmitting a diagnosis signal to the output end, and a second impedance varying circuit disposed on the second current path, and varying impedance of the second current path according to the first voltage or the second voltage.

(16) The second impedance varying circuit may adjust impedance of the second current path so that the second current maintains a first value according to the first voltage when the second current belongs to a first current section, and it may adjust impedance of the second current path so that the second current maintains a second value according to the second voltage when the second current belongs to a second current section. The first current section may be different from the second current section, and the first value may be different from the second value.

(17) The first impedance varying circuit may adjust impedance of the first current path so that the first current maintains the first value according to the first voltage when the first current belongs to a third current section. The first current section may be different from the third current section.

(18) The second impedance varying circuit may adjust impedance of the second current path according to the first voltage when the devices are normally connected, and it may adjust impedance of the second current path according to the second voltage when the interlock loop line is short-circuited to the ground.

(19) The first impedance varying circuit may maintain a turn-on state when the devices are normally connected, and it may adjust impedance of the first current path according to the first voltage when the interlock loop line is short-circuited to a positive electrode side of a battery.

(20) The first voltage varying circuit may include a first resistor disposed on the first current path, and a first shunt regulator element including an anode terminal and a reference terminal connected to respective ends of the first resistor and a cathode terminal for outputting the first voltage that is variable according to the first current flowing through the first resistor. The first impedance varying circuit may include a first transistor disposed on the first current path and including a first terminal connected to the input end, a second terminal connected to the first resistor, and a control terminal, and a second resistor connected between a cathode terminal of the first shunt regulator element and the control terminal of the first transistor. Impedance of the first transistor may be variable according to the first voltage.

(21) The second voltage varying circuit may include a third resistor disposed on the second current path, and a second shunt regulator element including an anode terminal and a reference terminal connected to respective ends of the third resistor, and a cathode terminal for outputting the second voltage that is variable according to the second current flowing through the third resistor. The second impedance varying circuit may include a second transistor disposed on the second current path, and including a first terminal connected to the node, a second terminal connected to the third resistor, and a control terminal connected to the cathode terminal of the first shunt regulator element and the cathode terminal of the second shunt regulator element, and a fourth resistor connected between the node and the control terminal of the second transistor. Impedance of the second transistor may be variable according to the first voltage or the second voltage. Resistance of the first resistor may be different from resistance of the third resistor.

(22) The connection diagnosis device may further include a passive element disposed on a current path between the input end and the ground, and having fixed impedance.

(23) The connection diagnosis device may further include a diagnosis unit for diagnosing connection states of the devices and identifying a short-circuit type of the interlock loop line based on at least one of a voltage and a current applied to the interlock loop line through the output end and a voltage and a current received through the input end.

(24) According to the embodiments of the present invention, when the short circuit is generated in

the interlock loop line, the generation of heat caused by the short circuit may be minimized and the circuit damages may be prevented. Further, it is possible to identify the short-circuit types when the short circuit is generated.

Description

DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 shows a connection diagnosis device according to an embodiment of the present invention.

(2) FIG. 2 shows an example of a circuit configuration of a connection diagnosis device according to an embodiment of the present invention.

(3) FIG. 3 shows a connection diagnosis device according to another embodiment of the present invention.

(4) FIG. 4 shows an example of a circuit configuration of a connection diagnosis device according to another embodiment of the present invention.

MODE FOR INVENTION

(5) Embodiments of the present invention will now be described in detail with accompanying drawings. Effects and characteristics of embodiments, and a realization method thereof, will now be described in detail with accompanying drawings. In the drawings, same reference numerals indicate same constituent elements, and no repeated descriptions thereof will be provided. As those skilled in the art would realize, the described embodiments may be modified in various different ways, all without departing from the spirit or scope of the present invention. The embodiments are provided as examples so that the present disclosure may be thorough and complete, and will sufficiently supply aspects and characteristics of the present invention to a person skilled in the art.

(6) Hence, for the purpose of complete understanding on the aspects and the characteristics of the present invention, processes, factors, and skills that may not be needed to a person of ordinary skill in the art may not be described. In the drawings, relative sizes of elements, layers, and regions may be exaggerated for clarity.

(7) In the present specification, the term “and/or” includes all or random combinations of a plurality of items that are related and arranged. When the embodiments of the present invention are described, the use of “may” signifies “at least one embodiment of the present invention”.

Regarding the description on an embodiment of the present invention, a singular term may include a plural form unless stated in another way.

(8) Terms including ordinal numbers such as first, second, and the like will be used only to describe various components, and are not to be interpreted as limiting these components. The terms are only used to differentiate one component from other components. For example, a first constituent element could be termed a second constituent element, and similarly, a second constituent element could be termed a first constituent element, without departing from the scope of the present invention.

(9) In describing the embodiments, the expression “connection” means electrical connection. Electric connection of two constituent elements includes not only a case where the two constituent elements are directly connected, but also a case where the two constituent elements are connected through another constituent element interposed therebetween. Other constituent elements may include a switch, a resistor, a capacitor, and the like.

(10) A connection diagnosis device according to embodiments of the present invention will now be described with reference to accompanying drawings.

(11) FIG. 1 shows a connection diagnosis device according to an embodiment of the present invention.

(12) Referring to FIG. 1, the connection diagnosis device **100** according to an embodiment of the

present invention may include a signal generator **110**, a plurality of voltage varying circuits **120** and **130**, a plurality of impedance varying circuits **140** and **150**, a passive element **160**, and a diagnosis unit **170**.

(13) The signal generator **110** may generate a diagnosis signal applied to an interlock loop line **L1** in a device **20** so as to diagnose the connection state of the device **20**. The diagnosis signal may be a voltage signal or a current signal. The diagnosis signal may also be a square-wave signal with a specific frequency. The connection state of the device **20** may signify that the device **20** is connected to another device (e.g., a connector, a battery pack cover, etc.). The connection diagnosis device **100** may be included in another device combined to the device **20**, or may be provided as an additional device.

(14) The voltage varying circuit **120** may output a varied output voltage according to a current **I11** flowing through a current path **CP11** for transmitting the diagnosis signal to the interlock loop line **L1**, that is, the current path **CP11** between the signal generator **110** and an output end **OUT10** of the connection diagnosis device **100**.

(15) The voltage varying circuit **130** may output a varied output voltage according to a current **I12** flowing through a current path **CP12** on which the signal input from the interlock loop line **L1** is transmitted, that is, the current path **CP12** between an input end **IN10** of the connection diagnosis device **100** and the ground **GND**.

(16) The impedance varying circuit **140** may be positioned on the current path **CP11** and may configure part of the current path **CP11**. The impedance varying circuit **140** may vary impedance of the current path **CP11** according to an output voltage of the voltage varying circuit **120**.

(17) The impedance varying circuit **150** may be positioned on the current path **CP12** and may configure part of the current path **CP12**. The impedance varying circuit **150** may vary impedance of the current path **CP12** according to an output voltage of the voltage varying circuit **130**.

(18) The voltage varying circuit **120** and the impedance varying circuit **140** constant-current-control the current **I11** flowing through the current path **CP11**, and may be operated to maintain a value of the current **I11** flowing through the current path **CP11**. For example, when the current **I11** flowing through the current path **CP11** becomes greater than a setting value **IA1**, the voltage varying circuit **120** adjusts an output voltage, and the impedance varying circuit **140** increases the impedance of the current path **CP11** according to the adjusted output voltage of the voltage varying circuit **120** so the current **I11** flowing through the current path **CP11** may be returned to the setting value **IA1**.

(19) The voltage varying circuit **130** and the impedance varying circuit **150** constant-current-control the current **I12** flowing through the current path **CP12**, and may be operated to maintain a value of the current **I12** flowing through the current path **CP12**. For example, when the current **I12** flowing through the current path **CP12** becomes greater than a setting value **IB1**, the voltage varying circuit **130** adjusts an output voltage, and the impedance varying circuit **150** increases the impedance of the current path **CP12** according to the adjusted output voltage of the voltage varying circuit **130** so the current **I12** flowing through the current path **CP12** may be returned to the setting value **IB1**.

(20) When the device **20** is normally connected, the output end **OUT10** and the input end **IN10** of the connection diagnosis device **100** may be electrically connected to an input end **IN2** and an output end **OUT2** of the interlock loop line **L1** in the device **20**. In this state, the diagnosis signal generated by the signal generator **110** is applied to the interlock loop line **L1** through the current path **CP11** and the output end **OUT10**, and is input to the input end **IN10** of the connection diagnosis device **100** through the interlock loop line **L1** so it may be transmitted through the current path **CP12**. The signal generator **110** may generate a diagnosis signal by using a voltage supplied from a battery (not shown). In this case, a voltage level of the diagnosis signal generated by the signal generator **110** may be varied according to an output voltage of a battery (not shown).

(21) When the voltage level of the diagnosis signal is increased to be greater than a predetermined

voltage or is reduced to be less than the same while the device **20** is normally connected, the size of the currents **I11** and **I12** flowing through the current path **CP11**, the interlock loop line **L1**, and the current path **CP12** may be changed. As described, when the size of the current flowing through the interlock loop line **L1** is changed according to a variation of the voltage level of the diagnosis signal while the device **20** is normally connected, the interlock loop line **L1** may be constant-current-controlled by the above-noted voltage varying circuit **120** and the impedance varying circuit **140**, or the voltage varying circuit **130** and the impedance varying circuit **150**.

(22) The passive element **160** may be positioned on the current path **CP12** and may configure part of the current path **CP12**. The passive element **160** may maintain the voltage of the interlock loop line **L1** to be a constant voltage while the device **20** is normally connected.

(23) According to what is described in the above, while the device **20** is normally connected, the current flowing to the interlock loop line **L1** may be maintained to be a constant current by operation of the impedance varying circuit **140** and the voltage varying circuit **120**, or the impedance varying circuit **150** and the voltage varying circuit **130**. As the interlock loop line **L1** maintains to be a constant current as described, the current flowing through the passive element **160** is maintained at a constant value, and the voltage of the interlock loop line **L1** may be controlled in a constant voltage state by the fixed impedance of the passive element **160**.

(24) To avoid an unstable operation, the impedance varying circuit **140** and the voltage varying circuit **120** may be configured to perform an impedance adjusting operation in a current section that is different from the impedance varying circuit **150** and the voltage varying circuit **130**. For example, the voltage varying circuit **120** and the impedance varying circuit **140** may be compared with the voltage varying circuit **130** and the impedance varying circuit **150**, and may control the constant current by adjusting the impedance of the current path **CP11** in a lower current section (e.g., a current section that may be generated when the device **20** is normally connected). in this case, the voltage varying circuit **130** and the impedance varying circuit **150** may be compared with the voltage varying circuit **120** and the impedance varying circuit **140**, and may control the constant current by adjusting the impedance of the current path **CP12** in a higher current section (e.g., an overcurrent section generated by a short circuit). The current value **IA1** constant-current-controlled by the voltage varying circuit **120** and the impedance varying circuit **140** may be less than the current value **IB1** constant-current-controlled by the voltage varying circuit **130** and the impedance varying circuit **150**.

(25) When the device **20** is assembled, the interlock loop line **L1** may be short-circuited to the ground or the battery (not shown) because of misconnection of lines. The short-circuit may generate an overcurrent, and when the overcurrent flows to the connection diagnosis device **100**, excessive heat may be generated or the circuit may be damaged. Therefore, to minimize the generation of heat according to the short-circuit and prevent the circuit from being damaged, the connection diagnosis device **100** may control currents of the current paths **CP11** and **CP12** with the predetermined values **IA1** and **IB1** through the voltage varying circuit **120** and the impedance varying circuit **140**, or the voltage varying circuit **130** and the impedance varying circuit **150**.

(26) When the interlock loop line **L1** is short-circuited to the ground, a short-circuit current that is greater than the setting value **IA1** flows to the current path **CP11**. In this case, the voltage varying circuit **120** may adjust the output voltage corresponding to the short-circuit current, and the impedance varying circuit **140** may increase the impedance of the current path **CP11** according to the adjusted output voltage of the voltage varying circuit **120** to thus constant-current-control the current **I11** flowing through the current path **CP11** with the setting value **IA1**.

(27) When the interlock loop line **L1** is short-circuited to a positive electrode of the battery (not shown), the greater short-circuit current than the setting value **IB1** flows to the current path **CP12**. In this case, the voltage varying circuit **130** may adjust the output voltage corresponding to the short-circuit current, and the impedance varying circuit **150** may increase the impedance of the current path **CP12** according to the adjusted output voltage of the voltage varying circuit **130** to

thus constant-current-control the current **I12** flowing to the current path **CP12** with the setting value **IB1**.

(28) The diagnosis unit **170** may be electrically connected to the input end **IN10**, may measure the voltage and the current input through the input end **IN10**, and may diagnose the connection state of the device **20**.

(29) According to what is described in the above, when the device **20** is normally fastened, the current input to the input end **IN10** through the interlock loop line **L1** while the diagnosis signal is applied is constant-current-controlled with a predetermined value (e.g., **IA1**), and the voltage of the interlock loop line **L1** measured through the input end **IN10** is constant-voltage-controlled to thus maintain a constant value. Therefore, the diagnosis unit **170** may identify the device **20** to be normally connected when the voltage input through the input end **IN10** is a value that is set by a constant-voltage-control or when the current input through the input end **IN10** is a value (e.g., **IA1**) that is set by a constant-current-control.

(30) The diagnosis unit **170** may identify a malfunction type when the connection state of the device **20** is determined to be bad.

(31) When the connection of the device **20** is in an opened state, the electrical connection between the output end **OUT10** or the input end **IN10** of the connection diagnosis device **100** and the interlock loop line **L1** of the device **20** may be disconnected. When the voltage and the current at the input end **IN10** are measured in this state, the voltage at the input end **IN10** may be measured to be a ground voltage or a limited voltage and the current input through the input end **IN10** may be measured to be 0 even when the diagnosis signal is applied.

(32) When the interlock loop line **L1** is short-circuited to the ground, the input end **IN10** and the output end **OUT10** may be connected to the ground. When the voltage and the current at the input end **IN10** and the output end **OUT10** are measured, the voltages at the input end **IN10** and the output end **OUT10** may be measured to be the ground voltage while the diagnosis signal is applied. Further, the current input from the interlock loop line **L1** through the input end **IN10** while applying the diagnosis signal is 0, and the current applied to the interlock loop line **L1** through the output end **OUT10** may be measured to be the value **IA1** that is set by the constant-current-control of the voltage varying circuit **120** and the impedance varying circuit **140**.

(33) When the interlock loop line **L1** is short-circuited to the side of the positive electrode of the battery, the input end **IN10** and the output end **OUT10** may be connected to the side of the positive electrode of the battery. When the voltage and the current at the input end **IN10** and the output end **OUT10** are measured, the voltages at the input end **IN10** and the output end **OUT10** while applying the diagnosis signal may be measured to be a voltage that is similar to the output voltage of the battery. The current input from the interlock loop line **L1** through the input end **IN10** while applying the diagnosis signal is the value **IB1** that is set by the constant-current-control of the voltage varying circuit **130** and the impedance varying circuit **150**, and the current flowing through the output end **OUT10** may be 0.

(34) Hence, the diagnosis unit **170** may measure the voltage and the current input from the interlock loop line **L1** through the input end **IN10** while applying the diagnosis signal or the voltage and the current output to the interlock loop line **L1** through the output end **OUT10**, and may identify defect types into opened, ground short circuit, and battery short circuit based on the measured values.

(35) FIG. 2 shows an example of a circuit configuration of a connection diagnosis device **100** of FIG. 1.

(36) Referring to FIG. 2, the signal generator **110** generates a diagnosis signal **Vbat** by using the output voltage of the battery (not shown), and the diagnosis signal **Vbat** may be a voltage signal that has an output voltage level of the battery.

(37) The voltage varying circuit **120** is configured with a shunt regulator circuit, and may include a shunt regulator element **U120** and a resistor **R120**. The shunt regulator element **U120** may include

an anode terminal, a cathode terminal, and a reference terminal. A resistor R120 may be connected between the anode terminal and the reference terminal of the shunt regulator element U120, and the cathode terminal of the shunt regulator element U120 may be connected to a control terminal of a transistor Q140 configuring the impedance varying circuit 140. The resistor R120 is positioned on the current path CP11 for transmitting the diagnosis signal Vbat applied by the signal generator 110 to the output terminal OUT10, and respective terminals may be connected to the anode terminal and the reference terminal of the shunt regulator element U120.

(38) The shunt regulator element U120 may be operated to vary the output voltage output to the cathode terminal according to the current I11 flowing through the resistor R120. For example, the shunt regulator element U120 may be operated to increase the voltage at the cathode terminal when the current I11 flowing through the resistor R120 becomes greater than the setting value IA1.

(39) The voltage at the cathode terminal of the shunt regulator element U120 that is variable according to the current I11 flowing through the resistor R120 is transmitted to a control terminal of the transistor Q140 configuring the impedance varying circuit 140, and may be used to determine impedance of the transistor Q140.

(40) The voltage varying circuit 120 may further include at least one of capacitors C121, C122, and C123 so as to remove ripples from the voltage output to the cathode terminal of the shunt regulator element U120 and remove noise. For example, to remove ripples from the output voltage, the capacitor C121 may be connected between the cathode terminal and the reference terminal of the shunt regulator element U120, and the capacitor C122 may be connected between the cathode terminal and the anode terminal of the shunt regulator element U120. For another example, a capacitor C123 for removing noise may be connected between the anode terminal of the shunt regulator element U120 and the ground.

(41) The voltage varying circuit 120 may further include at least one of diodes D121 and D122 so as to prevent a reversal of current toward the impedance varying circuit 140. The diode D121 includes a cathode terminal connected to the cathode terminal of the shunt regulator element U120, and an anode terminal connected to the control terminal of the transistor Q140, and may prevent the current from flowing to the control terminal of the transistor Q140 from the cathode terminal of the shunt regulator element U120. The diode D122 includes a cathode terminal connected to the resistor R120, and an anode terminal connected to the output terminal of the transistor Q140, and may prevent the current from flowing to the output terminal of the transistor Q140 from the resistor R120.

(42) The impedance varying circuit 140 may include a transistor Q140 and a resistor R140. The transistor Q140 is positioned on the current path CP11, and may include an input terminal, an output terminal, and a control terminal. The input terminal of the transistor Q140 may be connected to a node n11 to which the diagnosis signal Vbat is applied by the signal generator 110, and the output terminal of the transistor Q140 may be connected to the resistor R120 of the voltage varying circuit 120. The control terminal of the transistor Q140 may be connected to the cathode terminal of the shunt regulator element U120 configuring the voltage varying circuit 120. The resistor R140 may be connected between the node n11 to which the diagnosis signal Vbat is applied by the signal generator 110 and the control terminal of the transistor Q140.

(43) Impedance of the transistor Q140 may be variable according to the voltage input to the control terminal. The transistor Q140 configures part of the current path CP11 so the change of the impedance of the transistor Q140 may lead to a change of impedance of the current path CP11.

(44) For example, with reference to FIG. 2, the transistor Q140 may be configured with an NPN transistor. In this case, the input terminal, the output terminal, and the control terminal of the transistor Q140 may respectively be a collector terminal, an emitter terminal, and a base terminal. The impedance of the NPN transistor Q140 may increase as the current input to the base terminal is reduced. When the voltage at the cathode terminal of the shunt regulator element U120 configuring the voltage varying circuit 120 increases, the voltage between respective ends of the resistor R140

is reduced, and the current input to the NPN transistor Q140 through the resistor R140 is reduced. Therefore, the NPN transistor Q140 may be operated so that the impedance thereof may increase as the voltage at the cathode terminal of the shunt regulator element U120 increases, and the impedance may be reduced as the voltage at the cathode terminal of the shunt regulator element U120 is reduced.

(45) The transistor Q140 may be substituted with other types of transistors, not the NPN transistor.

(46) The impedance varying circuit 140 may further include a Zener diode D140 connected between the control terminal of the transistor Q140 and the ground so as to control the voltage applied to the control terminal of the transistor Q140.

(47) The voltage varying circuit 130 is configured with a shunt regulator circuit, and may include a shunt regulator element U130 and a resistor R130. The shunt regulator element U130 may include an anode terminal, a cathode terminal, and a reference terminal. A resistor R130 may be connected between the anode terminal and the reference terminal of the shunt regulator element U120, and the cathode terminal of the shunt regulator element U130 may be connected to the control terminal of the transistor Q150 configuring the impedance varying circuit 150. The resistor R130 may be positioned on the current path CP12 connected between the input end IN10 and the ground, and respective terminals may be connected to the anode terminal and the reference terminal of the shunt regulator element U130.

(48) The shunt regulator element U130 may be operated to vary the output voltage that is output to the cathode terminal according to the current I12 flowing to the resistor R130. For example, the shunt regulator element U130 may be operated to increase the voltage at the cathode terminal when the current I12 flowing through the resistor R130 becomes greater than the setting value IB1.

(49) The voltage at the cathode terminal of the shunt regulator element U130 that is variable according to the current I12 flowing through the resistor R130 may be transmitted to the control terminal of the transistor Q150 configuring the impedance varying circuit 150, and may be used to determine the impedance of the transistor Q150.

(50) The voltage varying circuit 130 may further include at least one of the capacitors C131 and C132 so as to remove ripples from the voltage output to the cathode terminal of the shunt regulator element U130. For example, to remove ripples from the output voltage, the capacitor C131 may be connected between the cathode terminal and the reference terminal of the shunt regulator element U130, and the capacitor C132 may be connected between the cathode terminal and the anode terminal of the shunt regulator element U130.

(51) The impedance varying circuit 150 may include a transistor Q150 and a resistor R150. The transistor Q150 may be positioned on the current path CP12, and may include an input terminal, an output terminal, and a control terminal. The input terminal of the transistor Q150 may be connected to the input end IN10, and the output terminal of the transistor Q150 may be connected to the resistor R130 of the voltage varying circuit 130 through the passive element 160. The control terminal of the transistor Q150 may be connected to the cathode terminal of the shunt regulator element U130 configuring the voltage varying circuit 130. The resistor R150 may be connected between the node n11 to which the diagnosis signal Vbat is applied by the signal generator 110 and the control terminal of the transistor Q150.

(52) The impedance of the transistor Q150 may be variable according to the voltage input to the control terminal. The transistor Q150 configures part of the current path CP12 so the change of impedance of the transistor Q150 may lead to a change of impedance of the current path CP12.

(53) For example with reference to FIG. 2, the transistor Q150 may be configured with an NPN transistor. In this case, the input terminal and the output terminal of the transistor Q150 may be a collector terminal and an emitter terminal, and the control terminal of the transistor Q140 may be a base terminal. The impedance of the NPN transistor Q150 may increase when the current input to the base terminal is reduced. When the voltage at the cathode terminal of the shunt regulator element U130 configuring the voltage varying circuit 130 increases, the voltage at respective ends

of the resistor **R150** is reduced, and the current input to the base of the NPN transistor **Q150** through the resistor **R150** is reduced. Hence, the NPN transistor **Q150** may be operated so that the impedance thereof may increase as the voltage at the cathode terminal of the shunt regulator element **U130** increases and the impedance may be reduced as the voltage at the cathode terminal of the shunt regulator element **U130** is reduced.

(54) The transistor **Q150** may be substituted with other types of transistor, not the NPN transistor.

(55) The impedance varying circuit **150** may further include a Zener diode **D150** connected between the control terminal of the transistor **Q150** and the ground so as to control the voltage applied to the control terminal of the transistor **Q150**.

(56) The passive element **160** may include a resistor **R160** positioned on the current path **CP12**. FIG. 2 exemplifies that the passive element **160** is positioned between the voltage varying circuit **130** and the impedance varying circuit **150**, and the position of the passive element **160** may be changed. For example, the passive element **160** may be disposed between the input end **IN10** and the transistor **Q150** configuring the impedance varying circuit **150**. For example, the passive element **160** may be disposed between the resistor **R130** configuring the voltage varying circuit **130** and the ground.

(57) According to what is described in the above, one of the impedance varying circuits **140** and **150** may perform an impedance adjusting function in the normal connection state and the overcurrent state, and the other thereof may perform an impedance adjusting function in the overcurrent state. That is, one of the transistors **Q140** and **Q150** configuring the impedance varying circuits **140** and **150** may be operated so that its impedance may be adjusted in the normal connection state and the overcurrent state, and the other may be operated so that its impedance may be adjusted in the overcurrent state. For this purpose, resistance of the resistor **R120** of the voltage varying circuit **120** may be set to be different from resistance of the resistor **R130** of the voltage varying circuit **130**. For example, resistance of the resistor **R120** may be set to be greater than resistance of the resistor **R130**.

(58) In the normal connection state of the device **20**, the transistor (e.g., transistor **Q150**) not performing an impedance adjusting operation may maintain a turn-on state in which it has constant impedance.

(59) FIG. 3 shows a connection diagnosis device according to another embodiment of the present invention.

(60) Referring to FIG. 3, the connection diagnosis device **200** according to another embodiment of the present invention may include a signal generator **210**, a plurality of voltage varying circuits **220** and **230**, a plurality of impedance varying circuits **240** and **250**, a passive element **260**, and a diagnosis unit **270**.

(61) The signal generator **210** may generate a diagnosis signal applied to the interlock loop line **L1** in the device **20** so as to diagnose the connection of the device **20**. The diagnosis signal may be a voltage signal or a current signal. The diagnosis signal may also be a pulse signal with a specific pattern.

(62) The voltage varying circuit **220** may output the output voltage that is variable according to the current **I21** flowing through the current path **CP21** for transmitting the diagnosis signal to the interlock loop line **L1**, that is, the current path **CP21** between the signal generator **210** and the output end **OUT20** of the connection diagnosis device **200**.

(63) The voltage varying circuit **230** may output the output voltage that is variable according to the current **I22** flowing through the current path **CP22** on which the signal input from the interlock loop line **L1** is transmitted, that is, the current path **CP22** between the input end **IN20** of the connection diagnosis device **200** and the ground **GND**.

(64) The impedance varying circuit **240** is positioned on the current path **CP21** and configures part of the current path **CP21**, and may vary the impedance of the current path **CP21** according to the output voltage of the voltage varying circuit **220** or the output voltage of the voltage varying circuit

230.

(65) The impedance varying circuit **250** is positioned on the current path CP22 to configure part of the current path CP22, and may vary the impedance of the current path CP22 according to the output voltage of the voltage varying circuit **230**.

(66) The voltage varying circuit **230** may, while the device **20** is normally connected, control the impedance varying circuit **240** so that the currents I21 and I22 may maintain constant values flowing through the current paths CP21 and CP22. For example, when the current I22 flowing through the current path CP22 becomes greater than a predetermined value IB2 while the device **20** is normally connected, the voltage varying circuit **230** adjusts the output voltage, and the impedance varying circuit **240** increases the impedance of the current path CP21 according to the adjusted output voltage of the voltage varying circuit **230** to thus return the currents I21 and I22 flowing through the current paths CP21 and CP22 to the predetermined value IB2.

(67) The voltage varying circuit **220** may control the impedance varying circuit **240** so that the current I21 flowing through the current path CP21 may be maintained at a constant value while the interlock loop line L1 is short-circuited to the ground. When the interlock loop line L1 is short-circuited to the ground and the short-circuit current that is greater than a predetermined value IA2 flows through the current path CP21, the voltage varying circuit **220** may adjust the output voltage according to the short-circuit current, and the impedance varying circuit **240** may increase the impedance of the current path CP21 according to the adjusted output voltage of the voltage varying circuit **220** to thus return the current I21 flowing through the current path CP21 to a predetermined value IA2.

(68) As described above, the impedance varying circuit **240** may be operated to receive the output voltages of the voltage varying circuit **220** and the voltage varying circuit **230**, vary the impedance of the current path CP21 according to the output voltage of the voltage varying circuit **230** in the normal connection state, and vary the impedance of the current path CP21 according to the output voltage of the voltage varying circuit **220** in the ground short-circuit state. Therefore, the impedance varying circuit **240** may adjust the impedance of the current path CP21 so that the currents I21 and I22 flowing through the current paths CP21 and CP22 may maintain the setting value IB2 in the normal connection state, and it may adjust the impedance of the current path CP21 so that the current I21 flowing through the current path CP21 may maintain the setting value IA2 in the ground short-circuit state. The setting value IA2 may be greater than the setting value IB2.

(69) The voltage varying circuit **230** may control the impedance varying circuit **250** so that the current I22 flowing through the current path CP22 may maintain a constant value while the interlock loop line L1 is short-circuited to the positive electrode side of the battery (not shown). When the interlock loop line L1 is short-circuited to the positive electrode side of the battery and the short-circuit current that is greater than the setting value IB2 flows through the current path CP22, the voltage varying circuit **230** may adjust the output voltage according to the short-circuit current, and the impedance varying circuit **250** may increase the impedance of the current path CP22 according to the adjusted output voltage of the voltage varying circuit **230** to thus return the current I22 flowing through the current path CP22 to the setting value IB2.

(70) The passive element **260** is positioned on the current path CP22, and may maintain the voltage of the interlock loop line L1 while the device **20** is normally connected. According to what is described in the above, while the device **20** is normally connected, the current flowing to the interlock loop line L1 may be maintained to be in the constant current state according to the operations of the impedance varying circuit **240** and the voltage varying circuit **220**, or the impedance varying circuit **240** and the voltage varying circuit **230**. As described above, as the interlock loop line L1 maintains the constant current state, the current flowing through the passive element **260** may be maintained, and the voltage of the interlock loop line L1 may be controlled to be the constant voltage by the fixed impedance of the passive element **260**.

(71) The diagnosis unit **270** may be electrically connected to the input end IN20 to measure the

voltage and the current input through the input end IN20, and may diagnose the connection state of the device 20 based on the measured results.

(72) When the device 20 is normally fastened, and while the diagnosis signal is applied, the current input to the input end IN20 through the interlock loop line L1 is constant-current-controlled with the setting value IB2, and the voltage of the interlock loop line L1 measured through the input end IN20 is constant-voltage-controlled to maintain a predetermined value. Therefore, when the voltage input through the input end IN20 is a value set by the constant-voltage control and the current input through the input end IN20 is a value IB2 set by the constant-current control, the diagnosis unit 270 may identify that the device 20 is normally connected.

(73) When the connection state of the device 20 is determined to be a defect, the diagnosis unit 270 may identify a defect type.

(74) When the voltage and the current of the input end IN20 are measured while the connection of the device 20 is opened, and the diagnosis signal is then applied, the voltage of the input end IN20 may be measured to be a ground voltage or a limit voltage, and the current input through the input end IN20 may be measured to be 0.

(75) When the voltage and the current on the input end IN20 and the output end OUT20 are measured while the interlock loop line L1 is short-circuited to the ground, the voltages at the input end IN20 and the output end OUT20 while the diagnosis signal is applied may be measured to be a ground voltage. The current input from the interlock loop line L1 through the input end IN20 while the diagnosis signal is applied is 0, and the current applied to the interlock loop line L1 through the output end OUT20 may be measured as the value IA2 set by the constant-current control of the voltage varying circuit 220 and the impedance varying circuit 240.

(76) When the interlock loop line L1 is short-circuited to the positive electrode side of the battery, the input end IN20 and the output end OUT20 may be connected to the positive electrode side of the battery. Hence, when the voltage and the current on the input end IN20 and the output end OUT20 are measured in this state, the voltages at the input end IN20 and the output end OUT20 while the diagnosis signal is applied may be measured to be similar to the output voltage of the battery. The current input from the interlock loop line L1 through the input end IN20 while the diagnosis signal is applied may be the value IB2 set by the constant-current control of the voltage varying circuit 230 and the impedance varying circuit 250, and the current flowing through the output end OUT20 may be 0.

(77) Therefore, the diagnosis unit 270 may measure the voltage and the current input from the interlock loop line L1 through the input end IN20 or the voltage and the current output to the interlock loop line L1 through the output end OUT20 while the diagnosis signal is applied, and may identify the defect types as opened, ground short-circuit, and battery short-circuit based on at least one of the measured values.

(78) FIG. 4 shows an example of a circuit configuration of a connection diagnosis device 200 of FIG. 3.

(79) Referring to FIG. 4, the signal generator 210 generates the diagnosis signal Vbat by using the output voltage of the battery (not shown), and the diagnosis signal Vbat may be a voltage signal having an output voltage level of the battery.

(80) The voltage varying circuit 220 is configured with a shunt regulator circuit, and may include a shunt regulator element U220 and a resistor R220. A resistor R220 is connected between the anode terminal and the reference terminal of the shunt regulator element U220, and the cathode terminal of the shunt regulator element U220 may be connected to the control terminal of the transistor Q240 configuring the impedance varying circuit 240. The resistor R220 is coupled in series to the current path CP21 for transmitting the diagnosis signal Vbat applied by the signal generator 210 to the output terminal OUT20, and respective terminals may be connected to the anode terminal and the reference terminal of the shunt regulator element U220.

(81) The shunt regulator element U220 may be operated to vary the output voltage that is output to

the cathode terminal according to the current **I21** flowing to the resistor **R220**. The voltage at the cathode terminal of the shunt regulator element **U220** may be transmitted to the control terminal of the transistor **Q240** configuring the impedance varying circuit **240**, and may be used to determine the impedance of the transistor **Q240**.

(82) The voltage varying circuit **220** may further include at least one of capacitors **C221**, **C222**, and **C223** so as to remove ripples from the voltage output to the cathode terminal of the shunt regulator element **U220** and remove noise. For example, to remove ripples from the output voltage, the capacitor **C221** may be connected between the cathode terminal and the reference terminal of the shunt regulator element **U220**, and the capacitor **C222** may be connected between the cathode terminal and the anode terminal of the shunt regulator element **U220**. For example, the capacitor **C223** for removing noise may be connected between the anode terminal of the shunt regulator element **U220** and the ground.

(83) To prevent the reversal of current toward the impedance varying circuit **240**, the voltage varying circuit **220** may further include at least one of diodes **D221** and **D222**. The diode **D221** may include a cathode terminal connected to the cathode terminal of the shunt regulator element **U220** and an anode terminal connected to the control terminal of the transistor **Q240**, and may prevent the current from flowing to the control terminal of the transistor **Q240** from the cathode terminal of the shunt regulator element **U220**. The diode **D222** may include a cathode terminal connected to the resistor **R220** and an anode terminal connected to the output terminal of the transistor **Q240**, and may prevent the current from flowing to the output terminal of the transistor **Q240** from the resistor **R220**.

(84) The voltage varying circuit **230** may include a shunt regulator circuit, a shunt regulator element **U230**, and a resistor **R230**. The shunt regulator element **U230** may include an anode terminal, a cathode terminal, and a reference terminal. The resistor **R230** may be connected between the anode terminal and the reference terminal of the shunt regulator element **U230**, and the cathode terminal of the shunt regulator element **U230** may be connected to the control terminal of the transistor **Q250** configuring the impedance varying circuit **250**. The resistor **R230** may be coupled in series to the current path **CP22** connected between the input end **IN20** and the ground, and the respective terminals may be connected to the anode terminal and the reference terminal of the shunt regulator element **U230**.

(85) The shunt regulator element **U230** may be operated to vary the output voltage that is output to the cathode terminal according to the current **I22** flowing to the resistor **R230**. The voltage at the cathode terminal of the shunt regulator element **U230** that is variable according to the current **I22** flowing through the resistor **R230** may be transmitted to the impedance varying circuits **240** and **250**, and may be used to determine the impedance of the transistors **Q240** and **Q250** configuring the impedance varying circuits **240** and **250**.

(86) The voltage varying circuit **230** may further include at least one of capacitors **C231** and **C232** so as to remove ripples from the voltage output to the cathode terminal of the shunt regulator element **U230**. For example, to remove ripples from the output voltage, the capacitor **C231** may be connected between the cathode terminal and the reference terminal of the shunt regulator element **U230**, and the capacitor **C232** may be connected between the cathode terminal and the anode terminal of the shunt regulator element **U230**.

(87) The impedance varying circuit **240** may include a transistor **Q240** and at least one of resistors **R241** and **R242**. The transistor **Q240** may be positioned on the current path **CP21**, and may include an input terminal, an output terminal, and a control terminal. The input terminal of the transistor **Q240** may be connected to a node **n11** to which the diagnosis signal **Vbat** is applied by the signal generator **210**, and the output terminal of the transistor **Q240** may be connected to the resistor **R220** of the voltage varying circuit **220**. The control terminal of the transistor **Q240** may be connected to the cathode terminals of the shunt regulator elements **U220** and **U230** configuring the voltage varying circuits **220** and **230**. The resistors **R241** and **R242** may be connected between the node

n11 to which the diagnosis signal Vbat is applied by the signal generator **210** and the control terminal of the transistor **Q240**.

(88) The impedance of the transistor **Q240** may be variable according to the voltage input to the control terminal. The transistor **Q240** configures part of the current path **CP21** so the change of impedance of the transistor **Q240** may lead to the change of impedance of the current path **CP21**.

(89) For example with reference to FIG. 4, the transistor **Q240** may be configured with an NPN transistor. In this case, the input terminal, the output terminal, and the control terminal of the transistor **Q240** may respectively be a collector terminal, an emitter terminal, and a base terminal. The impedance of the NPN transistor **Q240** may increase as the current input to the base terminal is reduced. When the voltage input to the control terminal of the NPN transistor **Q240** increases, the voltage at respective ends of the resistors **R241** and **R242** is reduced, and the current input to the base of the NPN transistor **Q240** through the resistors **R241** and **R242** is reduced. Therefore, the NPN transistor **Q240** may be operated so that the impedance thereof may increase as the voltage applied to the control terminal becomes greater, and the impedance thereof may be reduced as the voltage applied to the control terminal is reduced.

(90) The transistor **Q240** may be substituted with other types of transistors, not the NPN transistor.

(91) The impedance varying circuit **240** may further include a Zener diode **D240** connected between the control terminal of the transistor **Q240** and the ground so as to limit the voltage applied to the control terminal of the transistor **Q240**.

(92) The impedance varying circuit **250** may include a transistor **Q250** and a resistor **R250**. The transistor **Q250** is positioned on the current path **CP22**, and may include an input terminal, an output terminal, and a control terminal. The input terminal of the transistor **Q250** may be connected to the input end **IN20**, and the output terminal of the transistor **Q250** may be connected to the resistor **R230** of the voltage varying circuit **230** through the passive element **260**. The control terminal of the transistor **Q250** may be connected to the cathode terminal of the shunt regulator element **U230** configuring the voltage varying circuit **230** through the resistor **R250**. The resistor **R250** may be connected between the cathode terminal of the shunt regulator element **U230** and the control terminal of the transistor **Q250**.

(93) The impedance of the transistor **Q250** may be variable according to the voltage at the cathode terminal of the shunt regulator element **U230**. As the transistor **Q250** configures part of the current path **CP22**, the change of impedance of the transistor **Q250** may lead the change of impedance of the current path **CP22**.

(94) For example with reference to FIG. 4, the transistor **Q250** may be configured with an NPN transistor. In this case, the input terminal and the output terminal of the transistor **Q250** may be a collector terminal and an emitter terminal, and the control terminal of the transistor **Q240** may be a base terminal. The impedance of the NPN transistor **Q250** may increase when the current input to the base terminal is reduced. When the voltage at the cathode terminal of the shunt regulator element **U230** configuring the voltage varying circuit **230** increases, the current input to the base of the NPN transistor **Q250** through the resistor **R250** is reduced. Hence, the NPN transistor **Q250** may be operated so that the impedance thereof may increase as the voltage at the cathode terminal of the shunt regulator element **U230** increases and the impedance may be reduced as the voltage at the cathode terminal of the shunt regulator element **U230** is reduced.

(95) The transistor **Q250** may be substituted with other types of transistors, not the NPN transistor.

(96) The passive element **260** may include a resistor **R260** positioned on the current path **CP22**.

FIG. 4 exemplifies that the passive element **260** is positioned between the voltage varying circuit **230** and the impedance varying circuit **250**, and the position of the passive element **160** may be changed. For example, the passive element **260** may be disposed between the input end **IN20** and the transistor **Q250** configuring the impedance varying circuit **250**. For example, the passive element **260** may be disposed between the resistor **R230** configuring the voltage varying circuit **230** and the ground.

(97) According to what is described in the above, the voltage varying circuit **230** may be operated so that it may, when the device **20** is normally connected, control the impedance varying circuit **240** to adjust the impedance of the current path **CP21**, and the voltage varying circuit **220** may be operated so that it may, when the interlock loop line **L1** is ground-short-circuited, control the impedance varying circuit **240** to adjust the impedance of the current path **CP21**. That is, the voltage varying circuits **220** and **230** may control the impedance varying circuit **240** to adjust the impedance of the current path **CP21** in different current sections. To achieve this, regarding the circuit configuration of FIG. 4, resistance of the resistor **R220** of the voltage varying circuit **220** may be set to be different from resistance of the resistor **R230** of the voltage varying circuit **230**. The voltage varying circuit **220** may, while the short-circuit current flows through the current path **CP21**, control the impedance varying circuit **240**, and the voltage varying circuit **230** may control the impedance varying circuit **240** in the normal connection state so resistance of the resistor **R220** may be set to be less than resistance of the resistor **R230**.

(98) According to what is described in the above, the impedance varying circuit **250** may be operated to adjust the impedance of the current path **CP22** when an overcurrent flows to the current path **CP22** in a battery short-circuit condition, differing from the impedance varying circuit **240** for adjusting the impedance of the current path **CP21** in the normal connection state. For this operation, a resistor **R250** may be connected between the cathode terminal of the shunt regulator element **U230** and the control terminal of the transistor **Q250**. In the normal connection state of the device **20**, the transistor **Q250** may maintain the turn-on state having constant impedance.

(99) The connection diagnosis devices **100** and **200** according to the above-described embodiments may, when compared to the conventional HVIL circuits, minimize generation of heat according to a short circuit when the interlock loop line **L1** is short-circuited and may prevent circuit damages. In addition, the connection diagnosis devices **100** and **200** may control the interlock loop line **L1** in the constant current state and the constant voltage state, and may diagnose the connection and may identify defect types when a DC voltage signal is applied as well as a pulse width modulation (PWM) signal. Particularly, as the voltage of the interlock loop line **L1** is maintained at the constant voltage that is less than the output voltage of the battery in the normal state, and the voltage of the interlock loop line **L1** is maintained at the output voltage of the battery in the state that the interlock loop line **L1** is short-circuited to the battery, the connection diagnosis devices **100** and **200** may distinguish the normal state and the battery short-circuit state. On the contrary, when the DC voltage signal is applied to the interlock loop line in the existing HVIL circuits, the voltage of the interlock loop line in the normal state is similar to the voltage of the interlock loop line in the battery short-circuit state, so it is difficult to distinguish them, and the PWM signal is used as the diagnosis signal.

(100) Electronic or electrical devices according to embodiments of the present invention and/or other related devices or constituent elements may be realized by using appropriate hardware, firmware (e.g., an application-specific integrated circuit), software, or combinations of software, firmware, and hardware. For example, various configurations of the above-noted devices may be positioned on one integrated circuit (IC) chip or an individual IC chip. Various configurations of the above-noted devices may be realized on a flexible printed circuit film, a tape carrier package (TCP), a printed circuit board (PCB), or one substrate. The electrical or mutual connections described in the present specification may, for example, be realized by the PCB, wires on different types of circuit carriers, or conductive elements. The conductive elements may, for example, include metallization such as surface metallization and/or pins, and may include conductive polymers or ceramics. Electrical energy may be transmitted by electromagnetic radiation or a light-using radio access.

(101) The various configurations of the devices may be performed by at least one processor so as to perform the above-described various functions, they may be performed in at least one computing device, and they may be processes or threads for performing computer program instructions and

interacting with other system constituent elements. The computer program instruction is stored in a memory realizable in a computing device using a standard memory device such as a random access memory (RAM). The computer program instruction may also be stored in a non-transitory computer readable medium such as a CD-ROM or a flash drive.

(102) A person of ordinary skill in the art must understand that various functions of the computing device may be combined or united to a single computing device, or functions of a specific computing device may be dispersed to at least another computing device while not digressing from the range of the embodiments of the present invention.

DESCRIPTION OF SYMBOLS

(103) **100, 200**: connection diagnosis device **110, 210**: signal generator **120, 130, 220, 230**: voltage varying circuit **140, 150, 240, 250**: impedance varying circuit **160, 260**: passive element **170, 270**: diagnosis unit **CP11, CP12, CP21, CP22**: current path **IN10, IN20**: input end **OUT10, OUT20**: output end

Claims

1. A connection diagnosis device for diagnosing a connection state between devices, comprising: an output end and an input end connected to respective ends of an interlock loop line included in one of the devices; a first voltage varying circuit for outputting a first voltage that is variable according to a first current flowing through a first current path from among a current path for transmitting a diagnosis signal to the output end and a current path between the input end and a ground; and a first impedance varying circuit disposed on the first current path and varying impedance of the first current path according to the first voltage, wherein the first voltage varying circuit and the first impedance varying circuit are operated to maintain the first current at a constant value, wherein the first current path is the current path between the input end and the ground, wherein the connection diagnosis device further includes: a second voltage varying circuit for outputting a second voltage that is variable according to a second current flowing to a second current path for transmitting the diagnosis signal to the output end; and a second impedance varying circuit disposed on the second current path and varying impedance of the second current path according to the second voltage, wherein the second voltage varying circuit and the second impedance varying circuit are operated so that the second current maintains a constant value, wherein the first voltage varying circuit and the first impedance varying circuit are operated so that the first current maintains a first value, wherein the second voltage varying circuit and the second impedance varying circuit are operated so that the second current maintains a second value, and wherein the first value is different from the second value.
2. The connection diagnosis device of claim 1, wherein the first voltage varying circuit and the first impedance varying circuit are operated so that the first current maintains the first value when the first current belongs to a first current section, the second voltage varying circuit and the second impedance varying circuit are operated so that the second current maintains the second value when the second current belongs to a second current section, and the first current section is different from the second current section.
3. The connection diagnosis device of claim 1, wherein the second voltage varying circuit and the second impedance varying circuit are operated so that the second current maintains the second value while the devices are normally connected.
4. The connection diagnosis device of claim 1, wherein the first voltage varying circuit and the first impedance varying circuit are operated so that the first current maintains the first value while the interlock loop line is short-circuited to a positive electrode side of a battery, and the second voltage varying circuit and the second impedance varying circuit are operated so that the second current maintains the second value while the interlock loop line is short-circuited to the ground.
5. The connection diagnosis device of claim 1, wherein the first voltage varying circuit includes a

first resistor disposed on the first current path, and a first shunt regulator element including an anode terminal and a reference terminal connected to respective ends of the first resistor, and a cathode terminal for outputting the first voltage that is variable according to the first current flowing to the first resistor, and the first impedance varying circuit includes a first transistor disposed on the first current path, and including a first terminal connected to the input end, a second terminal connected to the first resistor, and a control terminal connected to a cathode terminal of the first shunt regulator element, and a second resistor connected between a node to which the diagnosis signal is input and the control terminal of the first transistor, and impedance of the first transistor is variable according to the first voltage.

6. The connection diagnosis device of claim 5, wherein the second voltage varying circuit includes a third resistor disposed on the second current path, and a second shunt regulator element including an anode terminal and a reference terminal connected to respective ends of the third resistor, and a cathode terminal for outputting the second voltage that is variable according to the second current flowing to the third resistor, the second impedance varying circuit includes a second transistor disposed on the second current path, and including a first terminal connected to the node, a second terminal connected to the third resistor, and a control terminal connected to a cathode terminal of the second shunt regulator element, and a fourth resistor connected between the node and the control terminal of the second transistor, and impedance of the second transistor is variable according to the second voltage.

7. The connection diagnosis device of claim 6, wherein resistance of the first resistor is different from resistance of the third resistor.

8. The connection diagnosis device of claim 1, wherein the first voltage varying circuit includes a first resistor disposed on the first current path, and a first shunt regulator element including an anode terminal and a reference terminal connected to respective ends of the first resistor and a cathode terminal for outputting the first voltage that is variable according to the first current flowing through the first resistor, the first impedance varying circuit includes a first transistor disposed on the first current path and including a first terminal connected to the input end, a second terminal connected to the first resistor, and a control terminal, and a second resistor connected between a cathode terminal of the first shunt regulator element and the control terminal of the first transistor, and impedance of the first transistor is variable according to the first voltage.

9. The connection diagnosis device of claim 8, wherein the second voltage varying circuit includes a third resistor disposed on the second current path, and a second shunt regulator element including an anode terminal and a reference terminal connected to respective ends of the third resistor, and a cathode terminal for outputting the second voltage that is variable according to the second current flowing through the third resistor, the second impedance varying circuit includes a second transistor disposed on the second current path, and including a first terminal connected to a node, a second terminal connected to the third resistor, and a control terminal connected to the cathode terminal of the first shunt regulator element and the cathode terminal of the second shunt regulator element, and a fourth resistor connected between the node and the control terminal of the second transistor, and impedance of the second transistor is variable according to the first voltage or the second voltage.

10. The connection diagnosis device of claim 9, wherein resistance of the first resistor is different from resistance of the third resistor.

11. The connection diagnosis device of claim 1, further comprising a passive element disposed on a current path between the input end and the ground, and having fixed impedance.

12. The connection diagnosis device of claim 1, further comprising a diagnosis unit for diagnosing connection states of the devices and identifying a short-circuit type of the interlock loop line based on at least one of a voltage and a current applied to the interlock loop line through the output end and a voltage and a current received through the input end.

13. A connection diagnosis device for diagnosing a connection state between devices, comprising:

an output end and an input end connected to respective ends of an interlock loop line included in one of the devices; a first voltage varying circuit for outputting a first voltage that is variable according to a first current flowing through a first current path from among a current path for transmitting a diagnosis signal to the output end and a current path between the input end and a ground; and a first impedance varying circuit disposed on the first current path and varying impedance of the first current path according to the first voltage, wherein the first voltage varying circuit and the first impedance varying circuit are operated to maintain the first current at a constant value, wherein the first current path is the current path between the input end and the ground, and wherein the connection diagnosis device further includes: a second voltage varying circuit for outputting a second voltage that is variable according to a second current flowing to a second current path for transmitting the diagnosis signal to the output end, and a second impedance varying circuit disposed on the second current path, and varying impedance of the second current path according to the first voltage or the second voltage, wherein the second impedance varying circuit adjusts impedance of the second current path so that the second current maintains a first value according to the first voltage when the second current belongs to a first current section, and it adjusts impedance of the second current path so that the second current maintains a second value according to the second voltage when the second current belongs to a second current section, wherein the first current section is different from the second current section, and wherein the first value is different from the second value.

14. The connection diagnosis device of claim 13, wherein the first impedance varying circuit adjusts impedance of the first current path so that the first current maintains the first value according to the first voltage when the first current belongs to a third current section, and the first current section is different from the third current section.

15. The connection diagnosis device of claim 13, wherein the second impedance varying circuit adjusts impedance of the second current path according to the first voltage when the devices are normally connected, and it adjusts impedance of the second current path according to the second voltage when the interlock loop line is short-circuited to the ground.

16. The connection diagnosis device of claim 14, wherein the first impedance varying circuit maintains a turn-on state when the devices are normally connected, and it adjusts impedance of the first current path according to the first voltage when the interlock loop line is short-circuited to a positive electrode side of a battery.
